

DATA CLEANING REPORT

This report presents the data cleaning and validation activities carried out on a dataset containing 1,200 records and 14 columns. The objective was to ensure that the dataset was complete, consistent, correctly formatted, and ready for further analysis.

1. Overview of the Dataset

Indicator	Result
Total rows before cleaning	1,200
Total rows after cleaning	1,200
Total columns checked	14

2. Missing and Null Values

All 14 columns were checked for missing or null values. The only missing values identified were found in the CouponCode column, which contained 309 blank rows. These blanks were interpreted as orders for which no coupon was used. To preserve this meaning and avoid leaving empty cells, the blank values were replaced with the label "NOCOUPON". No other missing values were found in the remaining columns.

3. Duplicate Records

Check performed	Result
Exact duplicate rows	0
Duplicate OrderID	0
Duplicate order with same Customer + Date	0

No duplicate records were identified during the validation process. Therefore, no rows needed to be removed or modified because of duplication.

4. Data Format Corrections

Several formatting checks and corrections were performed to improve the consistency of the dataset. The date column was verified and standardized using the YYYY-MM-DD format. Relevant columns such as OrderID, CustomerID, ShippingAddress, PaymentMethod, and OrderStatus were converted to an appropriate text format to ensure consistent interpretation of their values.

The UnitPrice and TotalPrice fields were rounded to two decimal places to maintain currency consistency, and their formatting was changed to numeric. Finally, the TotalPrice values were validated against the relationship $\text{TotalPrice} = \text{Quantity} \times \text{UnitPrice}$ for all 1,200 records.

5. Conclusion

The data cleaning process confirmed that the dataset contains 1,200 valid records after cleaning, with no rows lost during the process. The only missing values were addressed by replacing blank CouponCode entries with "NOCOUPON". No duplicates were detected, and the main data types and numeric formats were standardized. The TotalPrice calculation was also verified across the entire

dataset. Overall, the cleaned dataset is consistent and prepared for subsequent analysis and reporting.